

AI Critique - HW05 Performance Testing

During this assignment, I found that AI was useful for explaining performance testing concepts, suggesting JMeter configurations, and helping troubleshoot errors, but its suggestions were not always completely reliable. One important limitation was that AI sometimes assumed that a performance test was successful when the requests were technically completed, without sufficiently considering whether the workload was realistic or whether response time, throughput, CPU, memory usage, and other performance indicators remained acceptable. For example, a 100% success rate does not automatically mean that an endurance test has demonstrated a good performance threshold; the system could still experience significant latency or resource degradation. AI also sometimes suggested configurations or conclusions based on general performance-testing principles without having complete knowledge of the actual backend behavior and assignment requirements. This happened because AI does not automatically observe the running application, JMeter results, or resource-monitoring data unless these are explicitly provided. Therefore, some issues required me to run the tests, inspect the results, and verify the application's actual behavior manually. I learned that AI should be treated as an assistant rather than an authoritative source of truth. Its explanations and suggestions are useful for generating ideas and accelerating troubleshooting, but they must be validated against the real application, test results, assignment requirements, and objective evidence. The main principle I learned is to combine AI assistance with independent verification: understand the recommendation, test it in the actual environment, examine the evidence, and only then accept the conclusion.